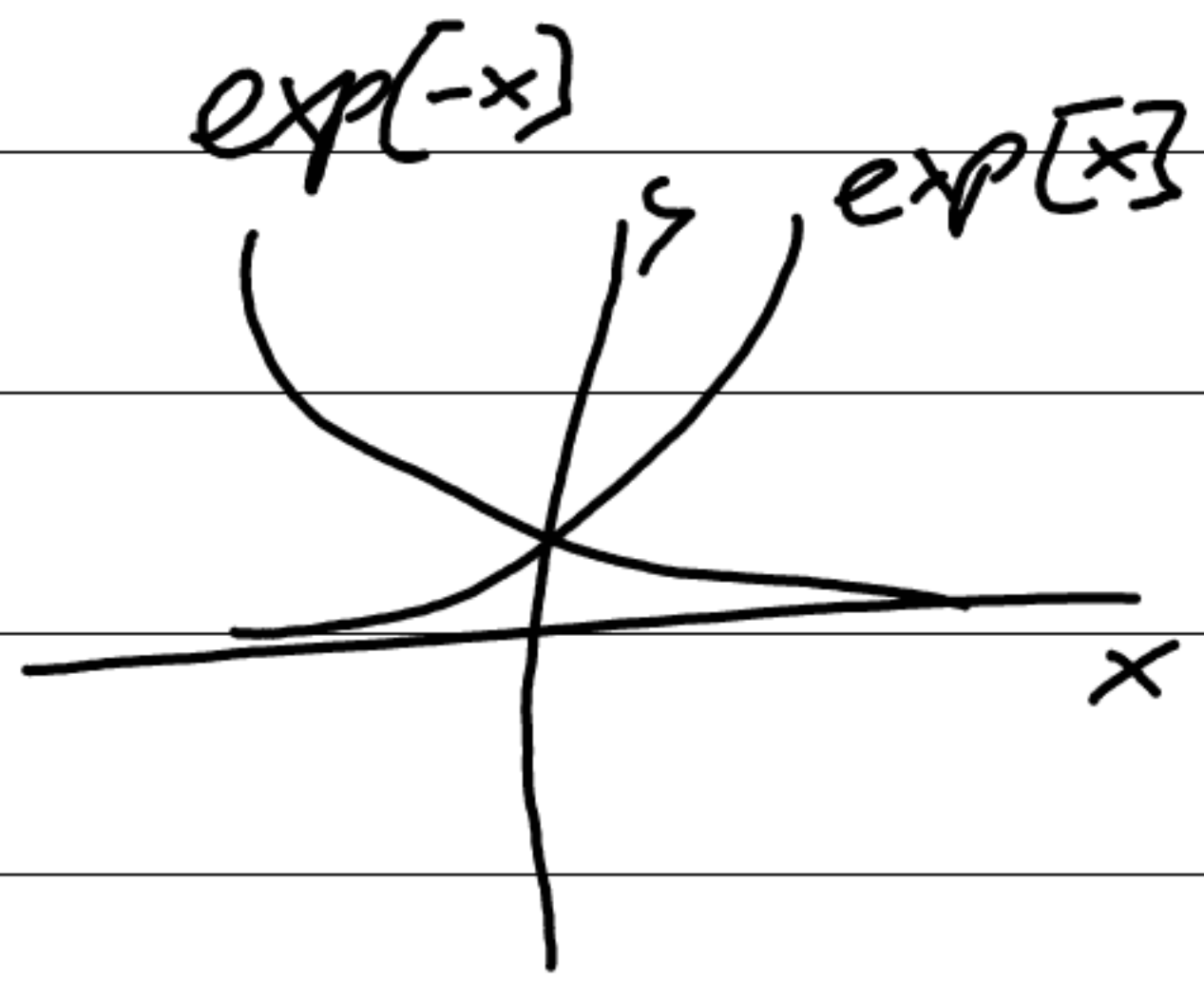


5 Loss functions

$$5.1 \rightsquigarrow 1) \lim_{z \rightarrow -\infty} \text{sig}[z] = 0$$

$$2) \text{sig}[0] = 0.5$$

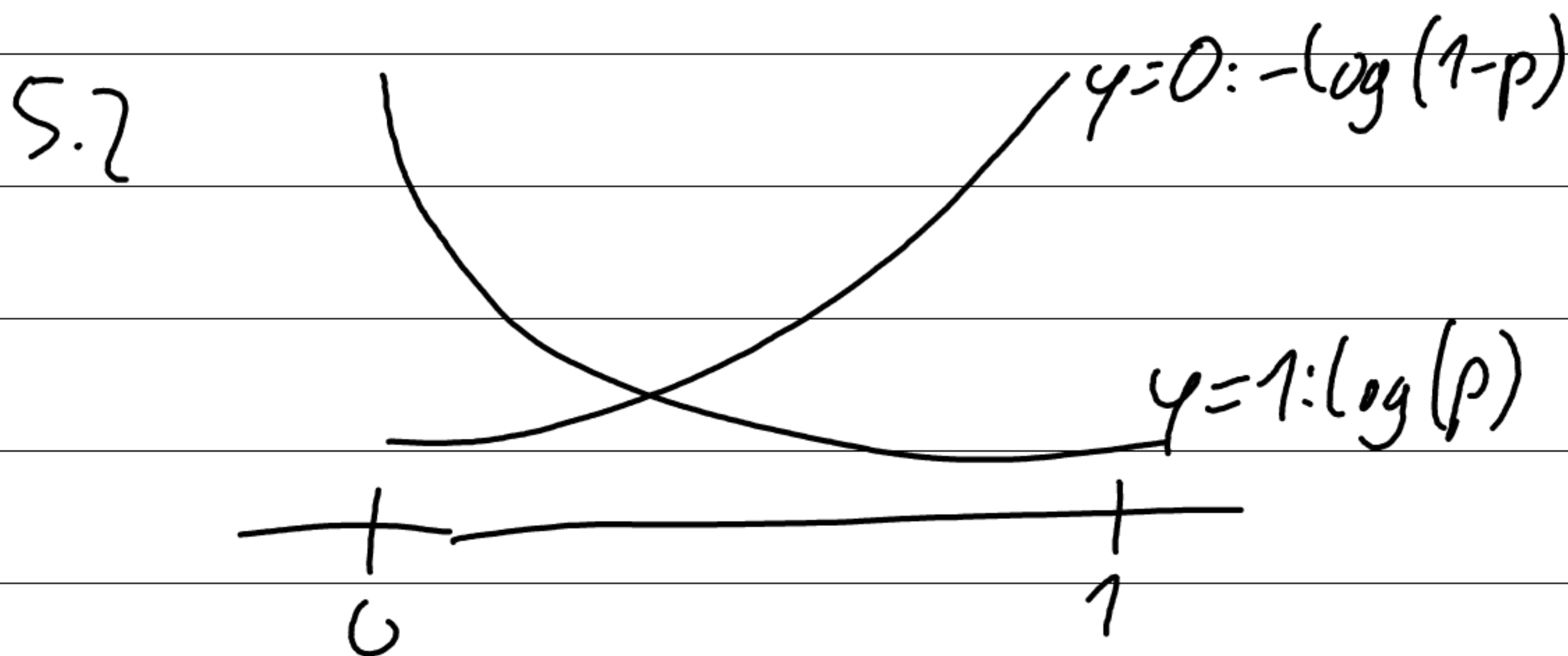
$$3) \lim_{z \rightarrow \infty} \text{sig}[z] = 1$$



$$\text{sig}[0.5] = \frac{1}{1 + \exp[0.5]} = \frac{1}{2}$$

$$\lim_{z \rightarrow -\infty} \text{sig}[z] = \frac{1}{1 + \lim_{z \rightarrow -\infty} \exp[z]} = 0$$

$$\lim_{z \rightarrow \infty} \text{sig}[z] = \frac{1}{1} = 1$$



$$5.3 \quad Pr(y|\mu, u) = \frac{\exp(i u \cos(y - \mu))}{2\pi \text{Bessel}_0[u]}$$

loss function for μ :

$$\mu = f(\bar{x}, \phi) \wedge Pr(y|\mu) = Pr(y|f(\bar{x}, \phi))$$

$$\phi = \underset{\phi}{\operatorname{argmin}} [L(\phi)] = \underset{\phi}{\operatorname{argmin}} \left[- \sum_{i=1}^n \log [Pr(y_i | f(\bar{x}_i, \phi))] \right]$$

$$L(\phi) = -u \sum_{i=1}^n \cos(y_i - f(\bar{x}_i, \phi)) \quad (\text{drop all terms constant in } \phi)$$

to perform inference, we can simply use our distribution $Pr(y|f(\bar{x}, \phi))$

$$5.4 \quad Pr(y, \lambda, \mu_1, \mu_2, \sigma_1^2, \sigma_2^2) = \frac{\lambda}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{(y-\mu_1)^2}{2\sigma_1^2}\right] + \frac{1-\lambda}{\sqrt{2\pi\sigma_2^2}} \exp\left[-\frac{(y-\mu_2)^2}{2\sigma_2^2}\right]$$

activation for params: $\lambda \in [0, 1] \leadsto \text{sigmoid: } \lambda_i = \text{sig}[f_1(\bar{x}_i, \phi)]$

$\mu_1, \mu_2 \in \mathbb{R} \leadsto \text{identity } \mu_{1,2} = f_{1,2}(\bar{x}_i, \phi)$

$\sigma_1^2, \sigma_2^2 \in \mathbb{R}^+ \leadsto \text{softplus } \log[1 + \exp[f_{2,3}(\bar{x}_i, \phi)]]$

$$\text{log-likelihood: } \sum_{i=1}^n \log \left[\frac{\lambda_i}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{(y_i - \mu_1)^2}{2\sigma_1^2}\right] + \frac{1-\lambda_i}{\sqrt{2\pi\sigma_2^2}} \exp\left[-\frac{(y_i - \mu_2)^2}{2\sigma_2^2}\right] \right]$$

$$L(\phi) = - \sum_{i=1}^n \log \left[\frac{\lambda_i}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{(y_i - \mu_1)^2}{2\sigma_1^2}\right] + \frac{1-\lambda_i}{\sqrt{2\pi\sigma_2^2}} \exp\left[-\frac{(y_i - \mu_2)^2}{2\sigma_2^2}\right] \right] \text{ where}$$

$\leadsto f(\bar{x}, \phi)$ would predict $(\lambda_i, \mu_{1i}, \mu_{2i}, \sigma_{1i}^2, \sigma_{2i}^2)$, we could use that for point estimates, sampling, etc.

$$\text{e.g., } E[Y|x] = \lambda \mu_1 + (1-\lambda) \mu_2$$

$$\text{Var}[Y|x] = E[Y^2|x] - E[Y|x]^2$$

$$= \lambda(\sigma_1^2 + \mu_1^2) + (1-\lambda)(\sigma_2^2 + \mu_2^2) - (\lambda \mu_1 + (1-\lambda) \mu_2)^2$$

$$= \lambda \sigma_1^2 + (1-\lambda) \sigma_2^2 + \lambda(1-\lambda)(\mu_1 - \mu_2)^2$$

↳ no closed-form solution for the node!

$$5.5 \quad Pr(\varphi | \lambda, \mu_1, \mu_2, \mu_1, \mu_2)$$

$$= \frac{\lambda \exp[\mu_1 \cos(\varphi - \mu_1)]}{2\pi \text{Bessel}_0[\mu_1]} + \frac{(1-\lambda) \exp[\mu_2 \cos(\varphi - \mu_2)]}{2\pi \text{Bessel}_0[\mu_2]}$$

$f(x, \phi) = (\lambda, \mu_1, \mu_2, \mu_1, \mu_2) \leadsto 5 \text{ parameters/outputs}$

$$5.6 \quad Pr(\varphi = k/\lambda) = \frac{\lambda^k \exp[-\lambda]}{k!}$$

$$L(\phi) = -\sum_{i=1}^I y_i \log(\lambda) - \lambda - \log(y_i!)$$

$$= \sum_{i=1}^I \exp[l(x_i, \phi)] - y_i l(x_i, \phi) + \log(y_i!)$$

$$\text{where } \lambda := \exp[l(x_i, \phi)]$$

5.7

independence!

$$Pr(y | f(\bar{x}, \phi)) = \prod_{d=1}^{10} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_d - \mu_d)^2}{2\sigma^2}\right]$$

$$\begin{aligned} \text{cl } |\Sigma| = |\sigma^2 I| &= 2\pi^{-D/2} |\Sigma|^{-1/2} \exp\left[-\frac{1}{2} (y - \mu)^T |\Sigma|^{-1} (y - \mu)\right] \\ &= (\sigma^2)^D \\ &= (2\pi\sigma^2)^{-5} \exp\left[-\frac{\|y - \mu\|^2}{2\sigma^2}\right] \\ &=: c \end{aligned}$$

$$MLL \mathcal{L}(\phi) = -\sum_{i=1}^1 (\log((2\pi\sigma^2)^5)) + -\frac{1}{2\sigma^2} \|y_i - f(x_i, \phi)\|^2, \text{ where } \mu = f(\bar{x}, b)$$

As c is constant, the Maximizing is equivalent:

$$\begin{aligned} \underset{\phi}{\text{argmax}} \mathcal{L}(\phi) &= -\sum_{i=1}^1 \overbrace{(\log((2\pi\sigma^2)^5))}^{\text{constant in } \phi} - \overbrace{\frac{1}{2\sigma^2} \|y_i - f(x_i, \phi)\|^2}^{\text{constant in } \phi} \\ &= \sum_{i=1}^1 \|y_i - f(x_i, \phi)\|^2 \quad \square \end{aligned}$$

$$5.8 \quad Pr(y | f(\bar{x}, \phi)) = \prod_{d=1}^{D_0} \frac{1}{\sqrt{2\pi\sigma_d^2}} \exp\left[-\frac{(y_d - \mu_d)^2}{2\sigma_d^2}\right]$$

$$MLL \mathcal{L}(\phi) = -\sum_{i=1}^1 \log(Pr(y | f(\bar{x}, \phi)))$$

$$\propto \sum_{i=1}^1 -\log(2\pi\sigma_{d_i}^2) + \frac{1}{\sigma_{d_i}^2} (y_{d_i} - \mu_{d_i})^2$$

$$\propto \sum_{i=1}^1 -\log(\sigma_{d_i}^2) + \frac{1}{\sigma_{d_i}^2} (y_{d_i} - \mu_{d_i})^2$$

$$= \sum_{i=1}^1 -\log(p_{2d}(\bar{x}, \phi)) + \frac{1}{p_{2d}(\bar{x}, \phi)} (y_{d_i} - p_{1d}(\bar{x}, \phi))^2$$

where $p_{2d}(\cdot) = \sigma_{d_i}^2$ or $p_{1d}(\cdot) = \mu_{d_i}$

5.9 $y = (\text{height}, \text{weight})$

Weight var is over much larger range than height

$\rightarrow$ weight errors dominate loss and backpropagation

solution: 1) standardize together, invert at inference

2) manually weight per-output (axes):

$$\lambda = \lambda_{\text{height}} \text{MSE}_{\text{height}} + \lambda_{\text{weight}} \text{MSE}_{\text{weight}}$$

5.10 $\Pr(y|\mu, \epsilon) = \frac{\exp[\epsilon \cos(\theta - \mu)]}{2\pi \text{Bessel}_0[\epsilon]}$ for direction

$$\Pr(s|m, \sigma) = \frac{1}{s\sigma\sqrt{2\pi}} \exp\left[-\frac{(\log(s) - m)^2}{2\sigma^2}\right], s > 0, \sigma > 0$$

Per speed (log-normal distribution).

Assuming independence: $\Pr(y, s | \mathcal{P}[x, \phi]) = \Pr(y|\mu, \epsilon) \cdot \Pr(s|m, \sigma)$ ^{assumed liked!}

$$\begin{aligned} \mathcal{L}(\phi) = & - \sum_{i=1}^I \log \left[\frac{\exp[\epsilon \cos(\theta_i - \mu_1[x, \phi])]}{2\pi \text{Bessel}_0[\epsilon]} \right. \\ & \left. \cdot \frac{1}{s_i \sigma_3[x, \phi]} \exp\left[-\frac{(\log(s_i) - \mu_2[x, \phi])^2}{2(\sigma_3[x, \phi])^2}\right] \right] \end{aligned}$$

where $\mu_1[x, \phi] = \mu$ (direction)

$\mu_2[x, \phi] = m$ (speed mean)

$\sigma_3[x, \phi] = \sigma$ (speed standard deviation)